SqFt Signal — Real Estate Investment Advisor: Project Documentation

Labmentix Internship · Project 7 · Business / Data Analyst batch

1. Executive summary

Out of 250,000 Indian property listings, 68,122 (27.2%) qualify as Good Investments under a transparent multi-factor rule, and every property is estimated to gain 27.63% in value over 5 years at 5% annual growth.

The project answers two investor questions: is this property a Good Investment, and what will it be worth after 5 years? The analysis cleaned and enriched the data, answered 20 EDA questions, ran a correlation analysis, and delivered the findings in a deployed Streamlit app, SqFt Signal (sqft-signal.streamlit.app).

The central finding is that no property feature drives price in this dataset. Size, location, BHK, age, amenities, schools, hospitals, transport, parking, furnishing and facing all show price gaps of 0.1–3.3%, and 88 of 91 numeric feature pairs are uncorrelated. The one reliable value signal is price per sq ft compared with the city median, which varies 3.3 times within each city. The dataset is synthetic, and every insight is reported with that limitation stated.

Headline result	Value
Properties analysed	250,000 (20 states, 42 cities)
Good Investments	68,122 (27.2%)
Perfect 5/5 Deal Score	3,006 (1.2%)
Median price per sq ft	₹9,245
Average price today → in 5 years	₹254.6 L → ₹324.9 L
Impossible floor values fixed	116,304 (46.5%)

2. Business problem and objectives

Investors need a fast, explainable way to separate good property deals from overpriced ones, and an estimate of future value.

Objectives

1. Classify each property as a Good Investment (Yes/No) using a clear, documented rule.
2. Estimate each property's value after 5 years.
3. Explain the market through 20 EDA questions and a correlation analysis.
4. Deliver the analysis in an interactive Streamlit app with filters, insights and an Investment Checker.

Business use cases

- Investors assess long-term returns before buying.
- Buyers find under-priced, high-quality properties.
- Real-estate platforms automate deal scoring for listings.
- Platforms build trust with data-backed, explainable ratings.

Approach decision: the project uses rule-based logic rather than machine-learning models, following the mentor's guidance that machine learning is optional for the Business / Data Analyst batch. A rule is fully transparent: every verdict can be explained criterion by criterion.

3. Dataset overview

The dataset (india_housing_prices.csv) holds 250,000 property listings and 23 columns, with no missing values and no duplicate rows.

Attribute	Value
Rows × columns	250,000 × 23 (49 after feature engineering)
Coverage	20 states, 42 cities, 500 localities
Property types	Apartment, Independent House, Villa (about 83,000 each)
Price	₹10–500 lakhs (mean ₹254.59 L, median ₹253.87 L)
Size	500–5,000 sq ft (mean 2,750, median 2,747)
BHK	1–5 (about 20% each)
Year built / age	1990–2023 / 2–35 years (age = 2025 − year built)
Nearby schools, hospitals	1–10 each
Categorical columns	Furnished status, public transport (Low/Medium/High), parking, security, facing, owner type, availability

The data is synthetic. Every numeric column is spread almost perfectly evenly (a uniform distribution): mean and median are nearly identical, and the standard deviation of price (141.35) matches the theoretical value for a uniform spread (141.45). Every category is split into near-equal groups (within 1%). Real markets are right-skewed and uneven, so all findings describe this dataset, not the actual Indian property market.

4. Data challenges

Nine data problems were found; each was fixed, worked around or documented, and no rows were removed.

#	Challenge	Impact	How it was handled
1	Floor_No > Total_Floors in 116,304 rows (46.5%)	Impossible floors; equal across all property types	Swapped the two values (e.g., floor 22 of 1 → floor 1 of 22) and flagged them in Floor_Fixed
2	Price_per_SqFt stored in lakhs, rounded to 2 decimals	Only 100 distinct values; minimum shows 0.00	Recalculated in rupees: Price × 1,00,000 ÷ Size (45,603 distinct values; max rounding error ₹500)
3	Amenities stored as text in random order	325 text variants of the same 5 items	Split into Amenities_Count (1–5) and five 1/0 flags
4	Locality names reused in all 42 cities	"Locality_84" is a code, not a neighbourhood	Analysed localities as codes (about 500 properties each) and stated the limitation
5	Villas and houses with up to 30 floors	Unrealistic building heights	Kept unchanged and documented; floor insights treated with caution
6	Parking_Space is Yes/No	The brief describes a number of spots	Compared with vs without parking
7	Owner_Type, Security, Availability labels differ from the brief	Brief: Individual/Agent, Gated/CCTV, Sold	Used the real labels (Owner/Builder/Broker, Yes/No, Ready_to_Move/Under_Construction)

#	Challenge	Impact	How it was handled
8	No future price, no Good Investment label	Required targets missing	Engineered both (Section 5)
9	No crime-rate or sale-date data	Brief questions on crime and price trends cannot be fully answered	Documented as a limitation; trends shown by construction period instead

The floor fix creates a correlation of +0.50 between floor and total floors (the floor is now always the smaller of two random numbers). This is a by-product of cleaning, not a market pattern.

5. Methodology

The work ran in eight phases in Google Colab, with the data loaded from the project's GitHub repository so the notebook runs end to end with Run all.

1. **Understand:** data types, summary statistics, unique values, category labels and the State → City → Locality structure.
2. **Clean:** confirmed 0 missing values and 0 duplicates (also with ID excluded); fixed impossible floors; checked age = 2025 − year built for all 250,000 rows.
3. **Preprocess:** Yes/No columns → 1/0 flags (Has_Parking, Has_Security, Is_Ready_To_Move); text → category types; ordered categories for transport (Low < Medium < High) and furnishing (Unfurnished < Semi-furnished < Furnished). Memory fell 35% (75.4 MB → 48.7 MB).
4. **Feature engineering:** 26 new columns (table below).
5. **EDA:** 20 questions, one chart each, with insight, implication and action (Section 6).
6. **Correlation analysis:** heatmap, drivers of the Good Investment label, and the brief's required pairs (Section 7).
7. **Save:** india_housing_cleaned.csv, 250,000 rows × 49 columns, reloaded and verified.
8. **App:** SqFt Signal, built in Streamlit and deployed on Streamlit Community Cloud (Section 9).

Key engineered features

Feature	Definition	Result
Price_per_SqFt_Rs	Price × 1,00,000 ÷ Size	Median ₹9,245; mean ₹13,058; range ₹202–₹99,182
Amenities_Count + 5 flags	Count and Yes/No for Gym, Pool, Garden, Clubhouse, Playground	1–5, about 20% each; each amenity in about 60% of homes
Infrastructure_Score	Schools, hospitals and transport each scaled to 0–1, averaged, × 10	0–10, mean 5.00; top 25% ≥ 6.5
Floor_Position	Floor ÷ total floors: Low ≤ 0.33, Mid ≤ 0.67, High above	34.6% / 33.3% / 32.1%
City_Median_PPSF, Price_vs_City_Median, Below_City_Median	Each property against its city's median ₹/sq ft	City medians ₹9,076–₹9,541 (5.1% gap)
Future_Price_5Y, Expected_Gain_Lakhs	Price × 1.05^5 (5% a year, compound)	+27.63% for every property; average gain ₹70.34 L
Investment_Score, Good_Investment	The rule below	68,122 Good Investments (27.2%)
Age_Group, Size_Band, Price_Band	Three bands each, for charts and filters	e.g., Small 22.3%, Medium 33.3%, Large 44.4%

The Good Investment rule

A property is a Good Investment when it is priced below its city median price per sq ft (the must-have) and meets at least 2 of 4 quality criteria.

Criterion	Condition	Share passing
Value (must-have)	Price per sq ft below the city median	50.0%
Ready to move	Availability = Ready_to_Move	50.0%
Strong infrastructure	Infrastructure Score ≥ 6.5	23.7%
Amenity-rich	4 or more amenities	40.1%
Security	Security available	50.1%

The Investment_Score counts the criteria met (0–5); a Good Investment needs the value test plus a score of at least 3. Transport is not scored separately because it is already one third of the Infrastructure Score.

Future price formula

Future_Price_5Y = Price × $(1 + 0.05)^5$ ≈ Price × 1.2763

6. EDA: answers to the 20 questions

Nineteen of the 20 questions show that the tested factor has no meaningful effect on price; the exceptions are mechanical (price per sq ft vs size) or created by the rule (investment potential by transport).

Each chart is in the notebook and in the app's Insight Explorer (repository folder assets/charts).

Price and size (Q1–Q5)

Q	Question	Chart	Answer	Investor action
1	Distribution of property prices	Histogram + mean/median lines	Even spread ₹10–500 L, about 5,000 per ₹10 L band; mean ₹254.59 L ≈ median ₹253.87 L; skewness 0.008	Judge value by price per sq ft vs the city median, not total price

Q	Question	Chart	Answer	Investor action
2	Distribution of property sizes	Histogram + mean/median lines	Even spread 500–5,000 sq ft, about 5,500 per 100 sq ft; mean 2,750 vs median 2,747; skewness 0.001	Choose size by target tenant; all sizes equally available
3	Price per sq ft by property type	Box plot	Medians: Independent House ₹9,305, Apartment ₹9,244, Villa ₹9,190 (1.3% gap)	Choose type by lifestyle, not price
4	Size vs price relationship	Scatter (5,000 sample) + trend line	No link (r = −0.003); larger homes are cheaper per sq ft (r = −0.615); Good Investment rate: Small 8.9%, Medium 22.4%, Large 40.1%	Large homes give the most space per rupee
5	Outliers in price per sq ft or size	Side-by-side box plots (IQR rule)	19,724 price-per-sq-ft outliers (7.9%) above ₹32,763; median 799 sq ft at ₹401 L; size has none	Avoid small homes above ₹32,763/sq ft; outliers kept, not deleted

Location (Q6–Q10)

Q	Question	Chart	Answer	Investor action
6	Average price per sq ft by state	Sorted horizontal bars	₹12,931 (Punjab) to ₹13,252 (Karnataka): 2.5% gap; median gap 3.3%	State choice barely changes the price paid

Q	Question	Chart	Answer	Investor action
7	Average property price by city	Sorted horizontal bars (42 cities)	₹250.8 L (Cuttack) to ₹258.5 L (Bangalore): 3.1% gap; Mumbai ₹252.1 L and New Delhi ₹251.6 L among the five cheapest	Pick a city for personal or rental reasons
8	Median property age by locality	Bars: localities per median age	99% of 500 localities at 17–20 years; 80% at exactly 18 or 19	Judge age property by property
9	BHK distribution across cities	Heatmap, diverging colours centred on 20%	Every BHK about 20% of every city; all 210 cells 18.6–22.0%	Every BHK type is available everywhere
10	Price trends, top 5 most expensive localities	Line chart by 5-year build period + benchmark	Top 5 average ₹269–275 L (6.7% above market) and swing randomly by period; overall line flat at ₹254–256 L	No locality is consistently premium; no sale dates, so not market growth

Feature relationships (Q11–Q15)

Q	Question	Chart	Answer	Investor action
11	Correlation between numeric features	Heatmap (lower triangle)	88 of 91 pairs show no link; 3 links are mechanical (Section 7)	Price per sq ft vs city median is the only value signal
12	Nearby schools vs price per sq ft	Box plot per count + median line	r = −0.000; medians ₹9,144–9,352 (2.3%)	School access comes free
13	Nearby hospitals vs price per sq ft	Box plot per count + median line	r = −0.000; medians ₹9,128–9,349 (2.4%)	Healthcare access comes free

Q	Question	Chart	Answer	Investor action
14	Price by furnished status	Bar chart of averages	Unfurnished ₹255.0 L, Semi ₹254.3 L, Furnished ₹254.4 L (0.3% gap)	Prefer furnished homes at no extra cost
15	Price per sq ft by facing	Bar chart of medians	North ₹9,219, East ₹9,234, South ₹9,225, West ₹9,308 (1.0% gap)	Vastu-preferred North/East cost no more

Investment, amenities and ownership (Q16–Q20)

Q	Question	Chart	Answer	Investor action
16	Properties per owner type	Bar chart, counts + shares	Broker 83,479, Owner 83,268, Builder 83,253 (33.3% each); Good Investment rate 27.0–27.4%	Buy direct from owners to avoid brokerage
17	Properties per availability status	Bar chart, counts + shares	124,965 ready vs 125,035 under construction; prices differ by 0.3%; Good Investment rate 38.4% vs 16.1% (rule-driven)	No under-construction discount: prefer ready-to-move
18	Does parking affect price?	Box plot, with vs without	₹254.7 L vs ₹254.4 L (+0.1%); median ₹/sq ft identical	Parking comes free
19	How amenities affect price per sq ft	Box plot by count + median line	r = 0.002; medians ₹9,207–9,294 (0.9%); single-amenity premiums −0.1% to +0.5%	Prefer amenity-rich homes at no extra cost

Q	Question	Chart	Answer	Investor action
20	Public transport vs price and investment potential	Two bar panels	Price flat (₹9,265 / ₹9,269 / ₹9,205 for Low / Medium / High); Good Investment rate 22.7% → 25.4% → 33.6%	High-transport homes are the best-value segment

Why median in some charts and mean in others: total price has no outliers, so averages are fair (Q14); price per sq ft is right-skewed with 19,724 outliers, so medians are used (Q3, Q15).

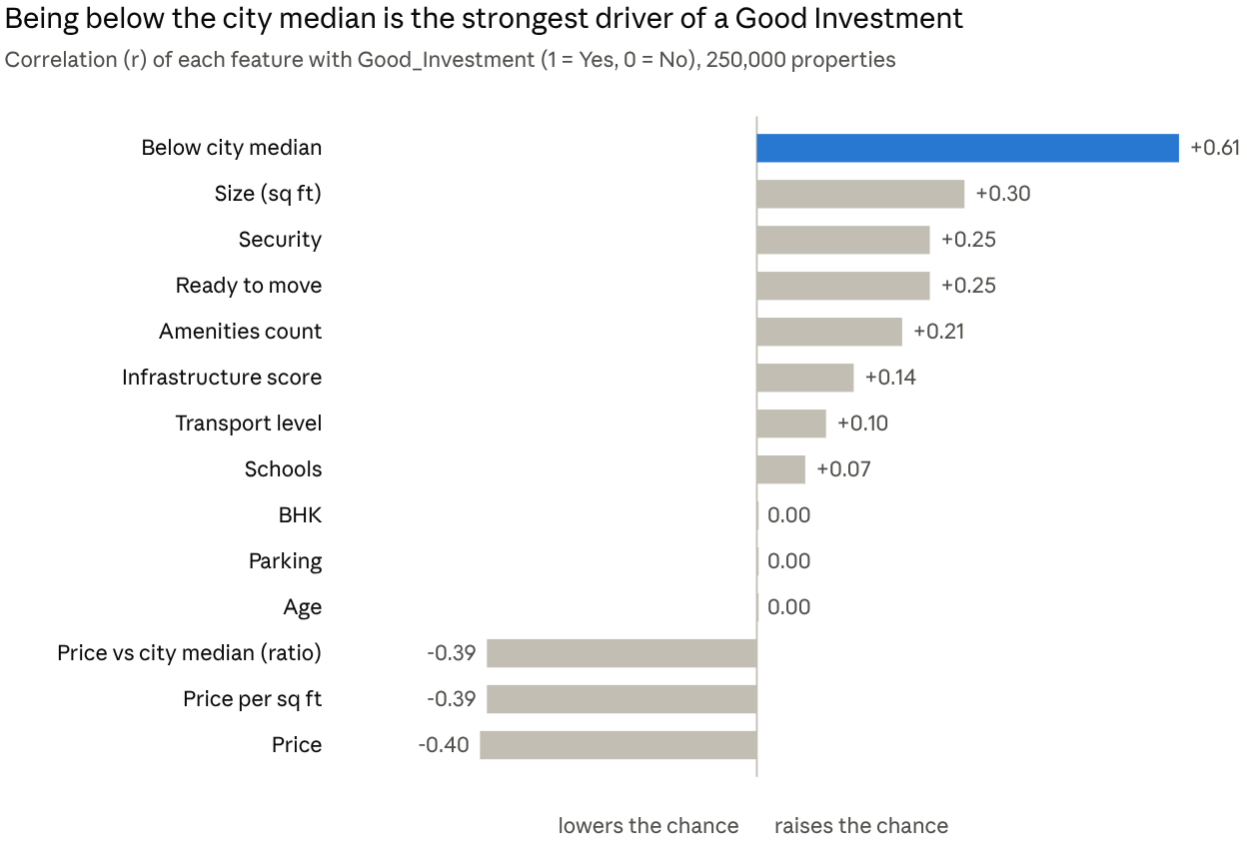
7. Correlation analysis

Being priced below the city median is by far the strongest driver of a Good Investment (r = +0.61), which confirms the rule's must-have works as designed.

Feature heatmap (Q11). Of 91 pairs among 14 numeric features, 88 have |r| below 0.05. The three correlated pairs are all mechanical:

- Size vs price per sq ft, r = −0.61: price per sq ft is price divided by size, and price does not rise with size.
- Price vs price per sq ft, r = +0.56: the same calculation.
- Floor vs total floors, r = +0.50: created by the cleaning swap.

Drivers of a Good Investment (Phase 6.1)



Notebook Phase 6.1 · correlations computed on all 250,000 properties

Size ranks second (+0.30) because large homes are cheaper per sq ft, so they pass the value test more often. Total price is the strongest negative (−0.40): cheaper homes tend to have lower price per sq ft. BHK, parking and age are not in the rule and show no link. The typical Good Investment is therefore a large, low-priced, ready-to-move home with security and amenities.

Pairs required by the project brief (Phase 6.2)

Pair	Result
Area (size) vs investment return	r = −0.003; median 5-year gain ₹70.8 L (small), ₹70.1 L (medium), ₹69.8 L (large)
Infrastructure score vs resale value	r = −0.001; median 5-year value ₹323.8 L (low), ₹324.3 L (medium), ₹323.5 L (high)
Crime rate vs Good Investment	Not possible: the dataset has no crime column

With one fixed growth rate, every property earns the same 27.6% return, so future value depends only on today's price, and today's price is unrelated to size and infrastructure.

8. Key insights

In this market, value comes from the individual property's price per sq ft, and quality features come at no extra cost.

1. **No feature drives price.** Every tested factor moves price by 0.1–3.3%, and 88 of 91 feature pairs are uncorrelated.
2. **Location barely matters.** States differ by 2.5% and cities by 3.1%; Mumbai and New Delhi are among the cheapest cities, the clearest sign the data is synthetic.
3. **Price per sq ft vs the city median is the real signal.** It varies 3.3 times within cities (25th percentile ratio 0.52, 75th 1.73) while cities differ by only 5.1%.
4. **Quality is free.** Amenities, schools, hospitals, transport, parking, furnishing and Vastu-preferred facing add no price premium.
5. **Large homes give the most space per rupee.** Price per sq ft falls with size ($r = -0.615$); 40.1% of large homes are Good Investments vs 8.9% of small.
6. **Ready-to-move beats under-construction.** There is no construction discount (0.3% price gap), so readiness removes risk at no cost.
7. **Overpriced outliers are small homes.** 19,724 properties (7.9%) sit above ₹32,763 per sq ft, with a median size of 799 sq ft.
8. **Every seller type offers the same deal quality.** Good Investment rates are 27.0–27.4% for owners, builders and brokers, so owner listings avoid brokerage at no quality cost.

9. Streamlit app: SqFt Signal

SqFt Signal is live at sqft-signal.streamlit.app and applies exactly the same rule and numbers as the notebook.

The name reflects the core finding: price per square foot against the city median is the one real signal in the data. The tagline is "Cut through the noise. Find the deals that matter."

Component	What it does
Sidebar filters	State → City (linked), property type, BHK, price and size sliders, live match counter, reset button
KPI cards	Properties, median ₹/sq ft, average price, Good Investment %, average 5-year value, each compared with the whole market

Component	What it does
Market Pulse	Four interactive Plotly charts that update with the filters: ₹/sq ft distribution, Deal Score distribution, top 10 cities by Good Investment rate, Good Investment rate by home size
Insight Explorer	All 20 EDA questions and 2 correlation analyses, each with its chart, finding, investor takeaway and chart rationale; Prev/Next navigation
Investment Checker	Enter any property to get a Good/Not Good verdict, a 0–5 Deal Score gauge, a value meter vs the city median, a ✅/❌ reason per criterion, a year-by-year 5-year value chart and a downloadable report
Top Picks	The best Good Investments in the filtered market as cards, ranked by Deal Score, then best value, lowest price or largest size
Recommendations	Evidence-backed advice for investors and platforms, plus a data reality check

Verification examples (Surat, city median ₹9,480/sq ft)

Test property	Result
3 BHK apartment, 3,000 sq ft, ₹200 L, ready, high transport, 8 schools, 7 hospitals, 5 amenities, security	Good Investment, 5/5; ₹6,667/sq ft (−29.7%); ₹255.3 L in 5 years
Same features at 2,400 sq ft and ₹300 L	Not Good, 4/5; ₹12,500/sq ft (+31.9%): the value test fails
4 BHK villa, 4,000 sq ft, ₹100 L, under construction, low transport, 1 amenity, no security	Not Good, 1/5; cheap (−73.6%) but meets 0 of 4 quality criteria

Build: Python, pandas, Plotly and Streamlit; custom styling (navy and teal theme, gradient banner, KPI cards, custom logo); data loaded once and cached; deployed from GitHub on Streamlit Community Cloud. Free apps sleep after inactivity; a visitor clicks the wake-up button and waits about 30 seconds.

10. Business recommendations

Investors should screen on value first, then collect the quality features that come at no extra cost; platforms should surface the same signal on every listing.

For investors

#	Recommendation	Evidence
1	Only buy below the city median price per sq ft	Strongest Good Investment driver (r = +0.61); price per sq ft varies 3.3× within a city
2	Favour larger homes	Price does not rise with size; 40.1% of large homes qualify vs 8.9% of small
3	Choose ready-to-move properties	Only 0.3% cheaper under construction; 38.4% of ready homes qualify vs 16.1%
4	Prioritise high public transport	Price flat across levels; 33.6% of high-access homes qualify vs 22.7% low
5	Collect free quality: amenities, schools, hospitals, parking, furnishing, East/North facing	None adds more than 0.5% to price per sq ft (schools and hospitals within 2.4%)
6	Avoid small homes above ₹32,763 per sq ft	19,724 outliers (7.9%), median 799 sq ft at ₹401 L
7	Buy direct from owners	Good Investment rate 27.0–27.4% for every seller type; brokerage is typically 1–2% of the price
8	Do not pay for a city or state name	Cities differ by 3.1% and states by 2.5%
9	Verify any listing priced 90% or more below its city median before acting	In a real market such prices usually signal data errors or misleading ads

For real-estate platforms

#	Recommendation	Evidence
1	Show a "% vs city median" badge on every listing	The clearest value signal found in 250,000 listings
2	Rank and flag listings with the 0–5 Deal Score	Only 3,006 listings (1.2%) score 5/5, so a top-pick badge stays exclusive

#	Recommendation	Evidence
3	Collect sale dates, RERA status, parking-spot counts and crime data	Needed for real price trends, location-based growth rates and risk scoring

11. Limitations and future work

The method is sound, but the synthetic data limits what the results say about the real Indian market.

Limitations

- The dataset is synthetic: values are uniform and no feature affects price.
- Good Investment is a transparent rule chosen by the analyst, not an observed market outcome; its thresholds (score ⩾ 6.5, 4+ amenities, 2 quality points) are judgement calls.
- The rule favours large homes, because they are cheaper per sq ft.
- Future value uses one fixed 5% growth rate, so every property gains the same 27.63%.
- No sale dates (no true price trends) and no crime data.
- Locality names are codes shared by all cities; villa and house floor counts are unrealistic.
- The analysis is not financial advice.

Future work

1. Use real historical prices to set growth rates by city and property type.
2. Add RERA, crime and sale-date data to score risk and track trends.
3. Train classification and regression models (Random Forest or XGBoost, tracked with MLflow) once real outcome data exists.
4. Add a size-adjusted benchmark (city median per size band) to reduce the large-home bias.

12. Conclusion and deliverables

The project turns 250,000 listings into a clear investment screen: 27.2% of properties qualify as Good Investments, each with an explainable reason and a 5-year value estimate, delivered in a live app.

The analysis also shows the importance of checking data honestly: the dataset's even spreads and missing price drivers were identified, measured and stated rather than hidden, and the rule's large-home bias was quantified.

Deliverable	Location
Google Colab notebook (all phases, text cells)	notebooks/Real_Estate_Investment_Advisor.ipynb in the GitHub repository
Cleaned dataset	india_housing_cleaned.csv (250,000 × 49), zipped for submission; compressed copy in the repository's data folder
Streamlit app	sqft-signal.streamlit.app; code (app.py, insights.py) in the repository
Charts	22 charts in the repository's assets/charts folder
Project documentation	This document
Explanatory video	7–10 minute walkthrough of the analysis and app